

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 2

2024

1. Let d be the metric on $\mathbb{C}$ defined by

$$d(z, w) = \begin{cases} 0 & \text{if } z = w \\ |z| + |w| & \text{if } z \neq w. \end{cases}$$

(This metric was introduced in Chp 2, Exercise 10. You should check at some stage that this really defines a metric on $\mathbb{C}$; but for now take that as given.)

- (a) Does the sequence (z_n) , with $z_n = \frac{1}{n} + \frac{1}{n}i$ converge to 0 in $(\mathbb{C}, d)$?
- (b) Does the sequence (z_n) , with $z_n = \frac{1}{n} + i$ converge to i in $(\mathbb{C}, d)$?
- (c) Characterise the convergent sequences in $(\mathbb{C}, d)$.

2. (a) Define, for non-empty subsets A, B of a set X ,

$$\text{dist}(A, B) = \inf\{d(a, b) \mid a \in A, b \in B\}.$$

Show that dist is *not* a metric on the collection of non-empty subsets of X . (Two properties fail - identify them both.)

- (b) Let (x_n) and (y_n) be sequences in a metric space (X, d) . We define (x_n) and (y_n) to be **neighbours** if $d(x_n, y_n) \rightarrow 0$ as $n \rightarrow \infty$. Suppose that $x_n \rightarrow a$ for some $a \in X$. Prove that the following conditions are equivalent:
 - i. (x_n) and (y_n) are neighbours
 - ii. $y_n \rightarrow a$
- (c) Prove that, for any non-empty subsets A, B of X , $\text{dist}(A, B) = 0$ iff there exist sequences (a_n) in A , (b_n) in B such that (a_n) and (b_n) are neighbours.

3. You are given that the function ρ defined by

$$\rho(x, y) = \begin{cases} 1 + |x - y| & \text{if one and only one of } x \text{ and } y \text{ is strictly positive} \\ |x - y| & \text{otherwise} \end{cases}$$

is a metric on $\mathbb{R}$. (You need not check this now.)

- (a) Decide whether the sequences $\left(-\frac{1}{n}\right)$ and $\left(\frac{1}{n}\right)$ converge in this space, and if so, to what limits.
- (b) Prove that this metric is *not* induced by a norm. (See pp 21/22 in the notes for how a norm induces a metric.)

4. Let

$$g_n(t) = \begin{cases} n(t - a) & \text{for } a \leq t \leq a + \frac{1}{n} \\ 1 & \text{for } a + \frac{1}{n} \leq t \leq b \end{cases}$$

and

$$g(t) = 1 \text{ for all } t \in [a, b]$$

- (a) Draw a graph of the function g_n .
- (b) Show that $g_n \rightarrow g$ with respect to the integral metric on $C([a, b])$, i.e. use $d(f, g) = \int_a^b |f(t) - g(t)| dt$. (If you look at your graph, you should be able to work this out without doing any integrating at all!)
- (c) Does $g_n \rightarrow g$ with respect to the supremum metric on $C([a, b])$, that is metric defined by $\sup\{|f(x) - g(x)| : x \in [a, b]\}$?
- (d) Prove or disprove: If $f_n \rightarrow f$ with respect to the integral metric on $C([a, b])$, then $f_n(x) \rightarrow f(x)$ for all $x \in [a, b]$.